



Webs, Hecke Characters, and Springer's Correspondence

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1 Diagrams

2 Varieties

3 Calculations

1 Diagrams Let $\mathbf{A} = \mathbf{Z}[a^{\pm 1}, q^{\pm 1/2}, \frac{1}{q^{1/2} - q^{-1/2}}]$.

The HOMFLYPT skein module of a surface S :

$$\text{Sk}(S) = \mathbf{A} \langle \mathcal{L}(S) \rangle / \text{Rel}(S).$$

$\mathcal{L}(S)$ is generated by oriented link diagrams.

$\text{Rel}(S)$ is generated by

$$\bigcirc = \frac{a - a^{-1}}{q^{1/2} - q^{-1/2}},$$

$$a \nearrow \searrow - a^{-1} \searrow \nearrow = (q^{1/2} - q^{-1/2}) \wr.$$

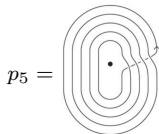
($\approx$ HOMFLYPT 1986) $\text{Sk}(\mathbf{R}^2) \simeq \mathbf{A}$.

Sends each oriented link to an $a, q^{1/2}$ -invariant.

How about $S = \mathbf{R}^2 - \mathbf{0}$?

Cannot simplify loops that go around $\mathbf{0}$.

Pairwise distinct diagrams p_k for $k \in \mathbf{Z}$.



Commutative ring structure via stacking diagrams.

(Turaev 1988) As an algebra,

$$\text{Sk}(\mathbf{R}^2 - \mathbf{0}) \simeq \mathbf{A}[p_{\pm 1}, p_{\pm 2}, p_{\pm 3}, \dots].$$

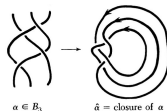
That is, an infinite-type polynomial algebra over $\mathbf{A}$.

Let Br_n be the braid group on n strands.

The diagrammatic *Hecke algebra* is

$$H_n = \frac{\mathbf{Z}[q^{\pm 1/2}][Br_n]}{\langle \nearrow - \nwarrow - (q^{1/2} - q^{-1/2}) \searrow \swarrow \rangle}.$$

The *annular closure* operation



defines a trace map

$$\text{ann}: H_n \rightarrow \text{Sk}(\mathbf{R}^2 - \mathbf{0}).$$

It factors through $\mathbf{A}[p_1, p_2, p_3, \dots]$.

(Murakami–Ohtsuki–Yamada 1998)

New generators for H_n from colored *webs*:

$$\begin{aligned} \left\langle \begin{array}{c} \uparrow \\ \swarrow \quad \searrow \\ \uparrow \end{array} \right\rangle_n &= q^{1/2} \left\langle \begin{array}{c} \uparrow \\ | \\ \uparrow \end{array} \right\rangle_n & \left\langle \begin{array}{c} \uparrow \\ | \\ \uparrow \end{array} \right\rangle_n &- \left\langle \begin{array}{c} \uparrow \quad \uparrow \\ \swarrow \quad \searrow \\ \uparrow \quad \uparrow \end{array} \right\rangle_n \\ \left\langle \begin{array}{c} \uparrow \\ \swarrow \quad \searrow \\ \uparrow \end{array} \right\rangle_n &= q^{-1/2} \left\langle \begin{array}{c} \uparrow \\ | \\ \uparrow \end{array} \right\rangle_n & \left\langle \begin{array}{c} \uparrow \\ | \\ \uparrow \end{array} \right\rangle_n &- \left\langle \begin{array}{c} \uparrow \quad \uparrow \\ \swarrow \quad \searrow \\ \uparrow \quad \uparrow \end{array} \right\rangle_n. \end{aligned}$$

The i th $\succcurlyeq$ generator: $C_i = q^{1/2} - \sigma_i = q^{-1/2} - \sigma_i^{-1}$.

In general webs, vertices are *trivalent* and obey *conservation of color*.

How to picture their annular closures?

$\mathbf{Z}[q^{\pm 1/2}]$ -linear combinations of *colored loops*.

Let e_n be a single loop with winding number 1 and color n .

$\mathbf{A}[p_1, p_2, p_3, \dots]$ is, secretly, the *ring of symmetric functions* over $\mathbf{A}$, with bases

$$\begin{aligned} \text{powersum} \quad \{p_\mu &= p_1^{\mu_1} p_2^{\mu_2} p_3^{\mu_3} \dots\}_\mu \\ \text{elementary} \quad \{e_\mu &= e_1^{\mu_1} e_2^{\mu_2} e_3^{\mu_3} \dots\}_\mu \end{aligned}$$

indexed by integer partitions μ .

Conj (Folklore) Any annular web

$$\text{ann}(C_{i_1} C_{i_2} \dots C_{i_\ell})$$

has a positive ($= \mathbf{Z}_{\geq 0}[q^{\pm 1/2}]$ -linear) expansion in $\{e_\mu\}_\mu$.

At the same time, the c_i are secretly elements of a *canonical basis* for H_n indexed by $z \in S_n$.

(Kazhdan–Lusztig 1979) Bases for H_n defined by

$$\begin{aligned} C_z &= \sum_{y \leq z} (-1)^{\ell_y} q^{(\ell_z - \ell_y)/2} P_{y,z}(q^{-1}) \sigma_y, \\ C'_z &= \sum_{y \leq z} q^{(\ell_y - \ell_z)/2} P_{y,z}(q) \sigma_y, \end{aligned}$$

where the polynomials are uniquely determined by $P_{z,z}(q) = 1$ and $\deg P_{y,z} \leq \frac{1}{2}(\ell_z - \ell_y - 1)$.

These bases are fixed by the ring anti-involution sending $q^{1/2} \mapsto q^{-1/2}$ and $\sigma_i \mapsto \sigma_i^{-1}$.

By contrast, the involution sending $q^{1/2} \mapsto q^{-1/2}$ and $\sigma_i \mapsto -\sigma_i$ swaps C_z with C'_z .

Under $H_n \xrightarrow{\text{ann}} \mathbf{A}[p_1, p_2, p_3, \dots]$, it swaps

$$\begin{aligned} &\text{elementary} \quad \{e_\mu = e_1^{\mu_1} e_2^{\mu_2} e_3^{\mu_3} \dots\}_\mu \\ &\text{with} \\ &\text{complete homogeneous} \quad \{h_\mu = h_1^{\mu_1} h_2^{\mu_2} h_3^{\mu_3} \dots\}_\mu. \end{aligned}$$

Conj ($\approx$ Haiman 1993) For any $z \in S_n$,

$$\text{ann}(C'_z) \quad (\text{resp. } \text{ann}(C_z))$$

has a positive expansion in the h_μ (resp. e_μ) for $\mu \vdash n$.

Subsumes the folklore conjecture.

In fact, Haiman's original formulation used characters, not diagrams.

The ring of symmetric functions is also

$$\bigoplus_{n \geq 1} \mathbf{A} \langle \text{Irr}(S_n) \rangle$$

under the product $\chi \cdot \psi := \text{Ind}_{S_n \times S_m}^{S_{n+m}} (\chi \boxtimes \psi)$.

$p_\mu \leftrightarrow$ rescaled indicator on cycle type μ'

$e_\mu \leftrightarrow \text{Ind}_{S_\mu}^{S_n}(\text{sgn})$

$h_\mu \leftrightarrow \text{Ind}_{S_\mu}^{S_n}(1)$

Schur $s_\mu \leftrightarrow$ irreducible χ^μ

What is annular closure in terms of characters?

For all $\beta \in H_n$,

$$\text{ann}(\beta) = \sum_{\lambda \vdash n} \chi_q^\lambda(\beta) s_\lambda,$$

where $\chi_q^\lambda \in \text{Irr}(H_n)$ deforms $\chi^\lambda \in \text{Irr}(S_n)$.

Conj (Haiman 1993) For any $z \in S_n$, if we expand

$$\text{ann}(C'_z) = \sum_{\mu \vdash n} \alpha_\mu^z h_\mu \quad \text{for some } \alpha_\mu^z \in \mathbf{Z}[q^{\pm 1/2}],$$

then the nonzero coefficients of α_μ^z are positive and unimodal for all μ .

Haiman checked this up through $n = 7$.

Why is this is hard?

The h_μ expand positively in the s_λ , but not vice versa.

Example For $z = s_2 s_1 s_3 s_2 \in S_4$,

$$\begin{aligned} & ([\mathbf{5}] + 3 \cdot [\mathbf{3}] + 2 \cdot [\mathbf{1}])_{s_4} \quad ([\mathbf{5}] + [\mathbf{3}])_{h_4} \\ \text{ann}(C'_z) = & + (2 \cdot [\mathbf{3}] + 2 \cdot [\mathbf{1}])_{s_{31}} = + [\mathbf{3}]_{h_{31}} \\ & + ([\mathbf{3}] + [\mathbf{1}])_{s_{22}} \quad + [\mathbf{3}]_{h_{22}} \end{aligned}$$

where $[\mathbf{m}] := \frac{q^{m/2} - q^{-m/2}}{q^{1/2} - q^{-1/2}}.$

(Hikita, Griffin–Mellit–Romero–Weigl–Wen 2025)

True for *312-avoiding* permutations z at $q^{1/2} \rightarrow 1$.

(The Stanley–Stembridge conjecture.)

How to generalize Haiman’s conjecture beyond S_n ?

It is about $\alpha_\mu^z \in \mathbf{Z}[q^{\pm 1/2}]$ defined by

$$\sum_{\chi \in \text{Irr}(S_n)} \chi_q(C'_z) \chi = \sum_{\mu \vdash n} \alpha_\mu^z \text{Ind}_{S_\mu}^{S_n}(1).$$

Can we generalize from S_n to, say, Weyl groups?

Issue 1 Beyond S_n , even the positivity of $\chi_q(C'_z)$ for irreducible χ can fail.

Issue 2 Beyond S_n , not enough conjugacy classes of parabolic subgroups to replace the S_μ .

We attempt to solve these issues through geometry.

2 Varieties Let G be a complex reductive group.

Let W be its Weyl group.

Let $B \subseteq G$ be a fixed Borel.

Springer theory tells us to build representations of W from the cohomology of the *Springer fibers*

$$\mathcal{B}_g = \{xB \in G/B \mid x^{-1}gx \in B\}.$$

Example Let G^{rs} be the regular semisimple locus.

If $G = \mathrm{GL}_n$, then $G^{rs} = \{\text{multiplicity-1 eigenvalues}\}.$

For any $g \in G^{rs}(\mathbf{C})$, the Springer fiber is a W -torsor.

(Springer 1978) For any $g \in G(\mathbf{C})$, still have

$$W \curvearrowright H^*(\mathcal{B}_g).$$

In fact, an injective map

$$\mathrm{Irr}(W) \hookrightarrow \left\{ (u, \kappa) \left| \begin{array}{l} u \in G(\mathbf{C}) \text{ unipotent,} \\ \kappa \in \mathrm{Irr}(\pi_0(G_u)) \end{array} \right. \right\} / \sim$$

where $\sim$ is conjugacy:

$$\chi \mapsto [u, \kappa] \iff \chi \text{ occurs in } H^{\mathrm{top}}(\mathcal{B}_u)_\kappa.$$

Example For $G = \mathrm{GL}_n$ and u unipotent of type μ :

$$\chi^\mu \text{ occurs in } H^*(\mathcal{B}_u) \simeq \mathrm{Ind}_{S_\mu}^{S_n}(1).$$

A generalization of Springer fibers:

For any $z \in W$, the *closed Lusztig variety*

$$\mathcal{Y}_{z,g} = \{xB \in G/B \mid x^{-1}gx \in \overline{BzB}\}.$$

For $g \in G^{rs}(\mathbf{C})$, monodromy of $\pi_1(G^{rs})$ descends to

$$W \curvearrowright \mathrm{IH}^*(\mathcal{Y}_{z,g}).$$

Thm (T. 2026, using Lusztig + Abreu–Nigro)

$$\sum_i q^{(i-\ell_z)/2} \mathrm{IH}^i(\mathcal{Y}_{z,g}) = \sum_{\chi, \psi \in \mathrm{Irr}(W)} \{\chi, \psi\} \psi_q(C'_z) \chi$$

where $\{-, -\}$ is Lusztig's exotic Fourier transform.

For $G = \mathrm{GL}_n$, r.h. sum is Haiman's $\sum_{\chi} \chi_q(C'_z) \chi$.

Issue 1 Beyond S_n , even the positivity of $\chi_q(C'_z)$ for irreducible χ can fail.

Issue 2 Beyond S_n , not enough conjugacy classes of parabolic subgroups to replace the S_{μ} .

Solutions (T., Brosnan–J. Hong–D. Lee–E. Lee 2026)

1 Replace $\sum_{\chi} \chi_q(C'_z) \chi$ with

$$\sum_{\chi, \psi \in \mathrm{Irr}(W)} \{\chi, \psi\} \psi_q(C'_z) \chi.$$

2 Replace the $\mathrm{Ind}_{S_{\mu}}^{S_n}(1)$ with

$$\textcolor{red}{spr}_{\psi} = H^*(\mathcal{B}_u)_{\kappa} \quad \text{for } \psi \in \mathrm{Irr}(W),$$

where (u, κ) is the Springer image of ψ .

3 Calculations So beyond type A:

Study the nonzero coefficients of the $\alpha_{\psi,G}^z \in \mathbf{Z}[q^{\pm 1/2}]$ defined by

$$\sum_{\chi, \psi \in \text{Irr}(W)} \{\chi, \psi\} \psi_q(C'_z) \chi = \sum_{\psi \in \text{Irr}(W)} \alpha_{\psi,G}^z \text{spr}_{\psi}.$$

Can use the GAP3 package CHEVIE to compute lots of examples.

! Although positivity often holds, it can still fail, starting in type B_2 where there is a single failure.

! Even up to absolute value, unimodality can fail, starting in type C_4 .

Example The $\alpha_{\psi,G}^z$ in type G_2 .

	e	2	1	21, 12	212	121
$\phi_{1,0}$				(101)	(1020)	(1020)
$\phi'_{1,3}$						
$\phi_{2,1}$				-1	-(10)	-(10)
$\phi_{2,2}$		(10)		1		(10)
$\phi'_{1,3}$			(10)	1	(10)	
$\phi_{1,6}$	1					

	2121, 1212	21212	12121	w_o
$\phi_{1,0}$	(10202)	(102020)	(102020)	(1020202)
$\phi'_{1,3}$	1	(10)	(10)	
$\phi_{2,1}$	-1			
$\phi_{2,2}$			-(10)	
$\phi'_{1,3}$	1		(10)	
$\phi_{1,6}$				

Above, $(a_m \cdots a_1 a_0) := a_0 + \sum_{i=1}^m a_i (q^{i/2} + q^{-i/2})$.

What properties of z or ψ ensure that $\alpha_{\psi,G}^z$ has positive unimodal coefficients?

Thm (T. 2026) For a Levi $G' \subseteq G$ with $W' \subseteq W$:

If $z \in W'$, then

the $\alpha_{\psi, G}^z$ for $\psi \in \text{Irr}(W)$

are $\mathbf{Z}_{\geq 0}$ -linear combinations of

the $\alpha_{\psi', G'}^z$ for $\psi' \in \text{Irr}(W')$.

Cor (T. 2026)

If $z \in W' \subseteq W$, and positivity and unimodality hold for the $\alpha_{\psi', G'}^z$, then they hold for the $\alpha_{\psi, G}^z$.

Proof

$\text{IH}^{2i}(\bar{Y}_{z, g})$ and spr_{ψ} enjoy the same compatibility with induction from W' to W .

Conj (T. 2026)

Suppose that $\psi \in \text{Irr}(W)$ is inflated from a product of symmetric groups.

Then for any $z \in W$, the nonzero coefficients of $\alpha_{\psi, G}^z$ all have the same sign and are unimodal up to sign.

If the inflation morphism does not descend to

$$W(G_2) \rightarrow W(A_2) = S_3,$$

then the sign is positive.

I checked this

- up through rank 5,
- in rank 6 for rationally smooth z ,
- in rank 7 for rationally smooth z of length ≤ 24 .

Example Type F_4 .

		unimodal	positive	inflated
$\phi_{1,0}$	F_4			✓
$\phi'_{2,4}$	$F_4(a_1)(11)$			✓
$\phi_{4,1}$	$F_4(a_1)$	×	×	
$\phi''_{2,4}$	$F_4(a_2)(11)$			✓
$\phi_{9,2}$	$F_4(a_2)$	×	×	
$\phi'_{8,3}$	C_3			
$\phi''_{8,3}$	B_3			
$\phi'_{1,12}$	$F_4(a_3)(211)$			✓
$\phi''_{6,6}$	$F_4(a_3)(22)$			
$\phi'_{9,6}$	$F_4(a_3)(31)$		×	
$\phi_{12,4}$	$F_4(a_3)$		×	
$\phi'_{4,7}$	$C_3(a_1)(11)$		×	
$\phi_{16,5}$	$C_3(a_1)$		×	
$\phi'_{6,6}$	$\tilde{A}_2 + A_1$		×	
$\phi_{4,8}$	$B_2(11)$			✓
$\phi''_{9,6}$	B_2		×	
$\phi''_{4,7}$	$A_2 + \tilde{A}_1$		×	
$\phi_{8,9}$	$\tilde{A}_2$			
$\phi'_{1,12}$	$A_2(11)$			✓
$\phi''_{8,9}$	A_2			
$\phi_{9,10}$	$A_1 + \tilde{A}_1$			
$\phi'_{2,16}$	$\tilde{A}_1(11)$			✓
$\phi_{4,13}$	$\tilde{A}_1$		×	
$\phi''_{2,16}$	A_1			✓
$\phi_{1,24}$	1			✓

Webpage with data and accompanying code: <https://mqtrinh.github.io/math/research/code/chevie/>

Claude Sonnet 4.6 helped me write the regex scripts.

Thank you for listening.